

Instructions:

- **Due February 9, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW02_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning objectives: This homework reinforces core control systems skills, including state-space modeling and linearization of nonlinear dynamical systems such as pendulums and robotic mechanisms.

Problem 1. Linearization. Write down the statement of [Taylor's theorem](#). Explain why Taylor's theorem is important for this course?

Problem 2. Linearization. Consider the simple pendulum

$$\ell\ddot{\theta} + g \sin \theta = 0. \quad (1)$$

1. Write the equations of motion in the state-space form.
2. Compute an equilibrium solution of the system.
3. Linearize the system about the equilibrium solution.

Problem 3. Linearization. Consider an inverted pendulum on a cart shown in Figure 1. The equations of motion are

$$(m + M)\ddot{x} + m\ell\ddot{\theta} \cos \theta - m\ell\dot{\theta}^2 \sin \theta = F, \quad (2)$$

$$(I + m\ell^2)\ddot{\theta} + m\ell\ddot{x} \cos \theta - mgl \sin \theta = 0. \quad (3)$$

1. Write the equations of motion in the state-space form.
2. Compute an equilibrium solution of the system.
3. Linearize the system about the equilibrium solution.

Hint: Define

$$\xi \triangleq \begin{bmatrix} x \\ \theta \end{bmatrix}. \quad (4)$$

Then, write the equation of motion in the form

$$\mathcal{M}(\xi)\ddot{\xi} + \mathcal{G}(\xi, \dot{\xi}) + \mathcal{K}(\xi) = \mathcal{F}. \quad (5)$$

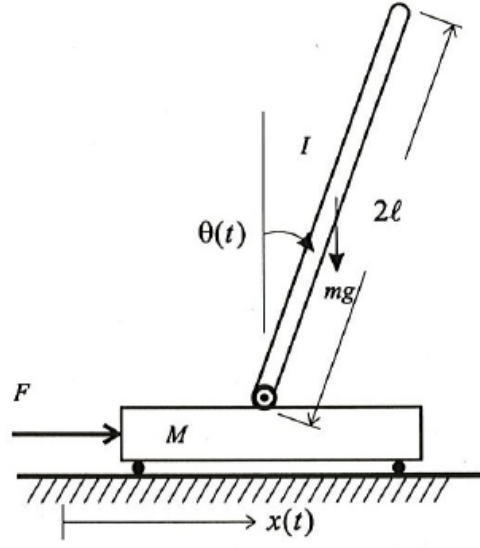


Figure 1: Inverted pendulum.

where $\mathcal{M}(\xi)$ is a 2×2 matrix, and $\mathcal{G}(\xi, \dot{\xi})$, $\mathcal{K}(\xi)$, $\mathcal{F}$ are 2×1 vectors constructed using the equations of motion. $\mathcal{M}(\xi)$ is called the mass matrix, $\mathcal{G}(\xi, \dot{\xi})$ is called the gyroscopic matrix, and $\mathcal{K}(\xi)$ is called the stiffness matrix. Then, define

$$\chi \triangleq \begin{bmatrix} \xi \\ \dot{\xi} \end{bmatrix}. \quad (6)$$

Then,

$$\dot{\chi} = \begin{bmatrix} \dot{\xi} \\ \mathcal{M}(\xi)^{-1} \left(\mathcal{F} - \mathcal{G}(\xi, \dot{\xi}) - \mathcal{K}(\xi) \right) \end{bmatrix} \quad (7)$$

is the state-space form. I recommend defining $\mathcal{M}, \mathcal{G}, \mathcal{K}, \mathcal{F}$ using symbolic variables in Matlab. You can then use the Matlab script [Automatic.Linearization.m](#) to compute the linearized dynamics.

Extra Credits Problem 1. 20 points. Linearization. Consider a robotic arm shown in Figure 2. The equations of motion are

$$(m_1 l_1^2 + I_1 + I_2 + m_2 d_2^2) \ddot{\theta}_1 + 2m_2 d_2 \dot{\theta}_1 \dot{d}_2 + (m_1 l_1 + m_2 d_2) g \cos \theta_1 = \tau_1, \quad (8)$$

$$m_2 \ddot{d}_2 - m_2 d_2 \dot{\theta}_1^2 + m_2 g \sin \theta_1 = \tau_2. \quad (9)$$

Note that m_1, m_2, I_1, I_2, g , and ℓ_1 are constants.

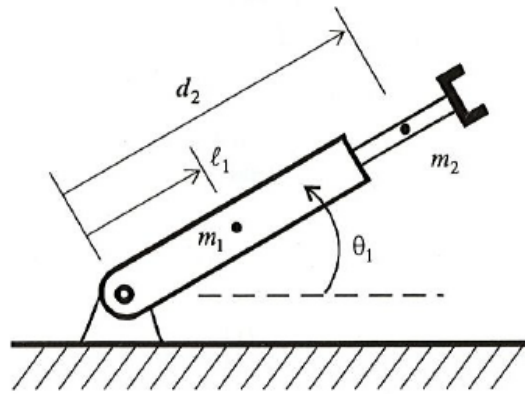


Figure 2: Robotic arm.

1. Write the equations of motion in the state-space form.
2. Compute an equilibrium solution of the system.
3. Linearize the system about the equilibrium solution.